

Exam Date & Time: 02-Jan-2025 (02:30 PM - 05:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

Manipal Institute of Technology, Bengaluru
BTech VII Semester - Make up Examination, Jan-2025

Cloud Computing [IT-4045]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) A company is considering migrating its IT infrastructure to a cloud platform. What key SLA considerations should they discuss with the cloud provider? How can a company ensure that their cloud service provider meets the agreed-upon SLAs? (5)
 - A)
 - B) A startup is looking to launch a new web application quickly and cost-effectively. How can SaaS help them achieve this goal? (3)
 - C) In a CIA Triad what do you mean by Availability? (2)
- 2) In a cloud computing environment, three users (User A, User B, and User C) are submitting tasks to a shared resource pool. The system uses a fair queueing algorithm to allocate resources. Each user submits tasks at different rates and requires different amounts of resources.
 - A)
 - User A submits tasks at a rate of 5 tasks per minute, each requiring 2 units of resource.
 - User B submits tasks at a rate of 3 tasks per minute, each requiring 3 units of resource.
 - User C submits tasks at a rate of 4 tasks per minute, each requiring 1 unit of resource.

The total available resource in the system is 20 units per minute. Based on the above scenario the following questions: (5)

 1. Calculate the total resource demand per minute for each user.
 2. Determine the fair share of resources each user should receive per minute based on their demand and the total available resources.
 3. Calculate the number of tasks each user can complete per minute with their fair share of resources.
 4. Evaluate the system's efficiency by comparing the total resource utilization with the available resources.
 - B) A company wants to use suitable desktop virtualization for supporting more number of users. Suggest a suitable desktop virtualization technique for them and explain. (3)
 - C) How can multi-factor authentication be used to improve security mechanisms? (2)
- 3) Virat as a chief technical officer(CTO) want to adapt Virtual Local Area Network(VLAN) in his enterprise. What are the benefits of VLAN and mention the steps for converting to configure the VLAN? (5)

- A)
- B) How site failure can be reduced in VDC environment. What are the steps could be adapted for Protection from Site Failure? (3)
- C) Mentioned how Demilitarized Zone is used to adapt security in the cloud environment. (2)
- 4) In a sports sector company, consider for a data set containing Name and Average_Goal key value pair, explain with a figure how would you proceed using Map-Reduce framework to find maximum scored player name? (5)
- A)
- B) Identify the different types of replica process adapted in the virtualized data centre. Explain. (3)
- C) A XYZ company wants to evaluate the performance and resource utilization of a proposed cloud infrastructure before deploying. What are the key steps involved in simulating this cloud environment using CloudSim? (2)
- 5) Imagine a global company with offices in multiple countries want to adapt to new Cloud Infrastructure Framework. Analyse and explain the various components used in it with a suitable diagram. (5)
- A)
- B) Provide the advantage of using Nested RAID - 0+1 in storage protection method (3)
- C) How to create a cloudlet in the cloudsim environment. (2)

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